Generalised Quantum Support Vector Machines for Cryptographic Classification

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Support vector machines (SVMs) are widely employed machine learning models for classification tasks but exhibit fundamental limitations when applied to datasets with complex, non-linear structures. In this research a generalised quantum support vector machine (QSVM) framework for cryptographic classification is presented, employing a fidelity-based quantum kernel that maps classical data into a high-dimensional Hilbert space via a parametrised quantum feature map. The proposed method employs simulated quantum circuits comprising single-qubit R_Z rotations and two-qubit R_{YY} entangling gates to encode cryptographically inspired datasets, derived from the Discrete Logarithm Problem for prime moduli in the range [11,373]. In comparison to classical SVMs with linear, polynomial, and radial basis function (RBF) kernels, the QSVM achieved mean accuracy improvements of 17.05%, 14.41%, and 8.65%, respectively, with all p-values < 0.05, thereby demonstrating statistically significant performance gains. An additional, brief analysis of circuit depth further suggested that the optimal quantum circuit depth may scale with dataset complexity up to a limit attributed to overfitting. These results indicate that quantum-enhanced kernels can capture intricate non-linear correlations beyond the reach of classical methods, offering promising avenues for cryptographic applications and potential extensions to domains such as genomics, materials science, finance, and climate modelling. Overall, this study lays a strong foundation for future hybrid quantum-classical architectures aimed at solving complex classification problems.

I. INTRODUCTION & BACKGROUND

Recent advancements in quantum information science and quantum computing have dramatically transformed our understanding of computational limits. Numerous studies suggest that problems previously considered intractable using classical computing can now be efficiently addressed using quantum algorithms [1]. Within this landscape, quantum machine learning (QML) has emerged as an especially interesting area to physicists, aiming to combine the high-dimensional power of quantum state spaces with the sophisticated data-analysis techniques of modern machine learning (ML) [2]. QML can refer to either the application of classical ML techniques to improve quantum systems (e.g. for error mitigation) or the integration of quantum processes and algorithms with classical ML methods [3]. This research explores the potential of the latter, particularly in the case of implementing Quantum Support Vector Machines (QSVMs).

Among supervised ML methods, Support Vector Machines (SVMs) have established themselves as a foundational model for classification tasks, having already demonstrated their capacity in tasks such as tumour classification for cancer diagnosis, email spam detection, and other standard ML tasks [4]. Yet, classical SVMs face intrinsic challenges when the interdependence of the input features becomes inherently complex. In cryptographic applications, datasets often arise from problems with conjectured exponential classical hardness, exemplified by the Discrete Logarithm Problem (DLP). Such datasets may exhibit subtle structure that can be extremely challenging for classical kernels to capture. These challenges naturally provide a reasonable environment to assess the potential for integration of quantum-enhanced methodologies into SVM architectures, setting the stage for the development of a generalised QSVM framework.

This research aims to develop, implement, and evaluate a generalised QSVM for cryptographic classification tasks, using DLP-derived datasets as a proof-of-concept. While the DLP-based classification serves as a test model, the framework proposed is intended to generalise to a wide range of real-world complex classification problems involving highly intricate pattern recognition ranging from personalised medicine via genomic classification to climate change forecasting and beyond.

The primary objectives of this research are to (i) develop a quantum kernel framework that unifies quantum mechanics with kernel-based learning, (ii) construct DLP-derived datasets with tunable complexity, (iii) implement parametrised quantum feature maps to encode classical data into multi-qubit states, (iv) compare QSVM performance with classical SVMs, and (v) optimise circuit depth for improved accuracy. The study ultimately seeks to determine whether QSVMs can consistently outperform classical kernels on challenging cryptographic datasets, thereby motivating their extension to real-world classification problems.

1. *Classical Support Vector Machines*

Classical SVMs form the baseline for this investigation. These models seek an optimal hyperplane maximising the separation between two data classes as shown in Figure 1.

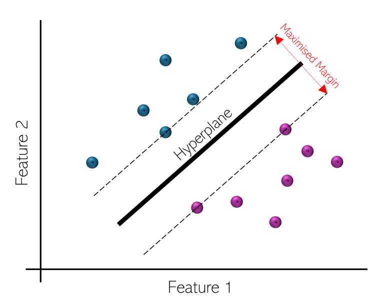


FIG. 1: An Illustration of how a classical SVM separates two classes.

For non-linearly separable data, a so-called "kernel trick" is employed to map the input data to a higher dimensional space where linear separation is feasible [4]. A simplified illustration of Radial Basis Function (RBF) kernel mapping is depicted in Figure 2 but other classical kernels are also common place in classical ML. This notion of kernel mapping underpins the "quantum-enhancement" implemented in this research. A full theoretical treatment of classical SVMs is detailed in Appendix A.

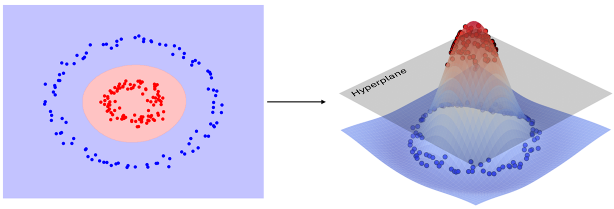


FIG. 2: Visualisation of the RBF kernel mapping, transforming non-linearly separable data into a higher-dimensional space.

2. *Quantum Computing for Machine Learning*

Quantum computing replaces classical bits with qubits that exist in superpositions of $|0\rangle$ and $|1\rangle$, where the state of n qubits resides in a 2^n-dimensional Hilbert space $\mathcal{H}_{2^n}$. Two key quantum phenomena underlie QML's potential advantages:

a. Superposition and Interference A qubit simultaneously holds amplitudes for $|0\rangle$ and $|1\rangle$, enabling n qubits to explore exponentially large state spaces. Quantum algorithms harness interference to amplify desired outcomes while suppressing others.

b. Entanglement A multi-qubit state $|\Psi\rangle$ is entangled if it cannot be factorised into individual qubit states. For a bipartite pure state, the Schmidt decomposition

$$|\Psi\rangle = \sum_{i=1}^{r} \sqrt{\lambda_i}\,|u_i\rangle \otimes |v_i\rangle \tag{1}$$

reveals its entanglement via the Schmidt coefficients $\{\lambda_i\}$; for mixed states, measures like concurrence and negativity quantify non-classical correlations. In QSVMs, entanglement is exploited to construct intricate, non-linear feature maps by coupling data features through entangled circuits, thereby mapping classical data into highly structured subspaces of $\mathcal{H}_{2^n}$ [5]. Moreover, each measurement in an entangled multi-qubit system collapses the global wavefunction, reflecting the intricate correlations that entanglement provides.

Classical input data $\mathbf{x}$ is first encoded into a quantum state $|\phi(\mathbf{x})\rangle$ via a unitary circuit $U_\phi(\mathbf{x})$. This circuit is composed of a variety of quantum operations performed using quantum gates including single-qubit rotations (e.g., $R_X(\theta)$, $R_Y(\theta)$, $R_Z(\theta)$) and entangling gates like the $R_{YY}(\theta)$ gate. Application of such a quantum circuit yields a quantum kernel defined as:

$$K_Q(\mathbf{x},\mathbf{y}) = \left|\langle \phi(\mathbf{x}) \mid \phi(\mathbf{y})\rangle\right|^2 = \left|\langle 0|U_\phi^\dagger(\mathbf{x})U_\phi(\mathbf{y})|0\rangle\right|^2 \tag{2}$$

This kernel, measuring the fidelity between quantum-encoded data points, leverages the exponential dimensionality of Hilbert space to reveal complex data correlations. The newly defined quantum kernel can be used analogously to classical kernels (including those described in the method and Appendix A) as it satisfies the properties of a kernel including symmetry, positive semi-definiteness and the ability to measure similarity between data points through an inner product in the transformed feature space [6]. As a result, the "quantum" SVM referenced throughout this work refers to a model that maps input data to a highly entangled Hilbert space before subsequently maximising class separation via the same optimisation process as the classical models. In principle, class separation may be more effective in this new space, increasing the efficacy of the model.

3. *The Discrete Logarithm Problem (DLP)*

The Discrete Logarithm Problem (DLP) is a foundational principle in public-key cryptography [7]. Here, consider a prime p and its multiplicative group $\mathbb{Z}_p^*$, whose elements are the integers $\{1, 2, \ldots, p-1\}$ under multiplication mod p. A generator g is an element that, through exponentiation, can produce every element of $\mathbb{Z}_p^*$. Formally, given (p, g, y), one seeks x such that

$$g^x \equiv y \pmod{p} \tag{3}$$

The problem is considered "one-way" because, while it is easy to compute y from x, recovering x from y is computationally hard.

Although not solving DLP itself, this research aims to use small p instances of it to create a cryptographically inspired classification task. This yields highly non-linear data structures, perfect for testing more advanced classification methods.

4. *Significance & Real-world Applications*

Although this study considers only a test dataset derived from the DLP, the same paradigm could be applied to global challenges where data complexity challenges classical methods including genomics [8], finance [9], and climate modelling [10] as discussed later in this report. By encoding data as quantum states, a QSVM aims to capture intricate correlations and latent structures in high-dimensional Hilbert spaces - features that classical kernels often overlook. The following sections detail the implementation and validation of this approach, laying the groundwork for future quantum-enhanced classifiers in a variety of critical, real-world applications.

II. METHODOLOGY

The research methodology for this work is divided into two phases. Firstly, a QSVM is implemented, trained and then tested with a fixed circuit depth and evaluated against classical kernels across several datasets (Phase I). Secondly, the effect of circuit depth on accuracy is explored, gradually increasing Reps until further gains in test accuracy saturate or regress as observed with hyperparameter tuning in classical machine learning (Phase II).

In general, both the quantum and classical models aim to learn a "decision boundary" that classifies unseen data points into one of two classes, $+1$ or -1. Each model is trained on datasets generated from simplified instances of the DLP for a variety of different primes $\{p\}$, and then evaluated on its ability to correctly predict the class labels of unseen data initially hidden from the model during the training phase.

The test dataset is generated by first, sampling exponents y within the multiplicative group $\mathbb{Z}_p^*$, computing residues $x = g^y \bmod p$, then mapping each point onto the unit circle:

$$x_{\sin} = \sin\left(\frac{2\pi x}{p}\right), \quad x_{\cos} = \cos\left(\frac{2\pi x}{p}\right) \tag{4}$$

Labels $\{\pm 1\}$ are assigned by comparing y to a secret value s. An example of one of these synthetic datasets is shown in Figure 3.

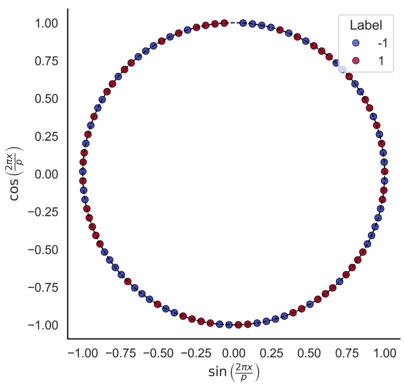


FIG. 3: Example DLP-derived dataset generated for $p = 101$ coloured by ± 1 classes.

By visual inspection, it is evident that there is no easily distinguishable pattern in the data class distribution. This characteristic is crucial for this study, since classical techniques have already demonstrated success in simple class separation, yet their efficacy in scenarios where separability is non-trivial remains a key area of exploration. The lack of an obvious structure ensures that the problem is sufficiently complex for testing quantum models. A significantly more detailed breakdown of the dataset construction is provided in Appendix B.

Before employing quantum kernels, three classical SVMs are implemented using `scikit-learn`'s `SVC`:

- **Linear Kernel:** $K_{\text{lin}}(\mathbf{x}, \mathbf{x}') = \mathbf{x} \cdot \mathbf{x}'$.
- **Polynomial Kernel:** $K_{\text{poly}}(\mathbf{x}, \mathbf{x}') = \left(\mathbf{x} \cdot \mathbf{x}' + c\right)^d$.
- **RBF Kernel:** $K_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \, \|\mathbf{x} - \mathbf{x}'\|^2\right)$.

Each classical kernel is tested 100 times on a dataset corresponding to each prime p (detailed in Appendix B) for confidence in its classification accuracy. In each run:

- Data generated by the prime is split into separate training (80%) and testing (20%) subsets.
- Each SVM is trained on the designated training data.
- The SVMs' performance is evaluated on the unseen test set to ensure a fair comparison of model accuracy.

From these 100 runs, a mean test accuracy and an empirical standard error is determined, thereby quantifying model variability more precisely.

In parallel to classical baselines, a Pauli Feature Map is constructed that embeds $\mathbf{x} = (x_{\sin}, x_{\cos})$ into a multi-qubit Hilbert space. The unitary

$$U_\Phi(\mathbf{x}) = \prod_{k=1}^{\text{Reps}} \Bigg[\underbrace{\prod_{i=1}^{D} R_Z(x_i)}_{\text{single-qubit } Z\text{-rotations}} \quad \underbrace{\prod_{i<j}^{D} R_{YY}(x_i, x_j)}_{\text{two-qubit entangling}} \Bigg] \tag{5}$$

encodes each feature (x_i) into the quantum state. This operator is selected as it can be considered a fairly generalised quantum embedding operator (with extended detail superposition and entangling gates in Appendix C). Here:

- Reps: The number of repeating blocks (often referred to as depth).
- D: Feature dimension and thus, number of qubits.
- $R_Z(x_i)$: Rotates qubit i about the Z-axis by an angle related to x_i.
- $R_{YY}(x_i, x_j)$: Applies a controlled entangling rotation between qubits i and j based on (x_i, x_j).

This design is intended to be highly generalised, not explicitly encoding the encryption operation, therefore building on previous research inspiring this work for example by UC Berkeley and IBM Quantum [6]. Instead, it systematically couples each feature to every other feature, leveraging superposition and entanglement to create a potentially high-capacity kernel.

The circuit configuration chosen for the first stage was a unitary quantum circuit with $Reps = 5$ (balancing expressivity with noise) and $D = 2$ (one qubit per classical feature). Diagrams and further details on the implemented circuits throughout this research can be audited in Appendix C.

After building the feature-map circuits $U_\phi(\mathbf{x})$ and $U_\phi(\mathbf{y})$, Qiskit's `FidelityQuantumKernel` numerically computes the overlap from Eq. (2) by combining them into $U_\phi^\dagger(\mathbf{x})\, U_\phi(\mathbf{y})$ and simulating its action on $|0\rangle^{\otimes n}$. In the statevector backend, this means evolving and then directly reading off the amplitude of the all-zero final state. Because this has to be done for every pair $(\mathbf{x}, \mathbf{y})$, constructing the full $N \times N$ kernel is significantly more expensive than a simple dot product, so the quantum model was limited to three runs per prime.

Phase 1: Fixed-Depth Analysis

In the first stage of this research, the quantum and classical models were evaluated on datasets derived from primes 20 different small primes in the range $[11, 373]$ as detailed in Appendix B. Comparison with classical baselines (linear, polynomial, and RBF SVMs) is thus made by comparing these averaged accuracies.

Phase 2: Circuit Depth Optimisation

The second phase investigates whether incrementally increasing the circuit depth (i.e., the number of feature-map repetitions Reps) yields higher accuracy for more complex classification tasks. This is analogous to hyperparameter tuning in classical SVMs: starting from $\text{Reps} = 1$, repeatedly build deeper circuits, recalculate training/test kernels, retrain the QSVM, and stop once the accuracy fails to improve further. Unfortunately, due to hardware constraints, this analysis was only performed up to a circuit depth of 5.

Although a thorough cross-validation of multiple Reps values was not computationally feasible here, applying the simple iterative approach described above allowed for a rough estimation of the "optimal depth" for a subset of tested primes: $\{59, 101, 373\}$. While deeper circuits could potentially offer richer feature spaces, they also risk overfitting (in simulation) and accumulate noise on real quantum hardware.

Specifics of the iterative depth-increment procedure, can be found in Appendix D whilst the full Python code implementation for this methodology is given in Appendix G.

III. RESULTS & DISCUSSION

The performance of the implemented QSVM was evaluated against classical SVMs employing linear, polynomial, and RBF kernels. The classification accuracy of each model was assessed on independent test datasets, with results averaged over multiple trials. Figure 4 compares mean accuracies across the total set of selected primes for the four models, illustrating that QSVM achieves higher overall accuracy than any classical SVM across the sampled prime values.

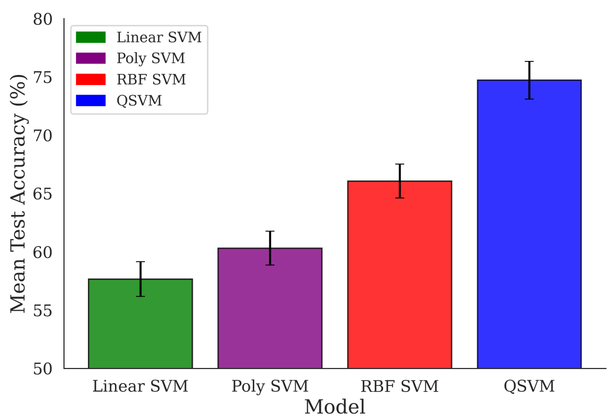


FIG. 4: Mean classification accuracy for each model, averaged over multiple trials. QSVM (right) outperforms all three classical models.

Table I summarises the results of two-sample t-tests comparing average test accuracy across the full set of primes of the QSVM against each classical SVM. The low p-values indicate that the QSVM's performance improvements are unlikely to arise from random fluctuations. Specifically, these results demonstrate that, across the selected set of primes, the QSVM outperforms each classical model with statistical significance as the largest p-value observed was just below 5%, while the other comparisons yielded considerably smaller p-values. Appendix E provides a comprehensive account of all numerical results as well as more detail on the statistical methodology behind this analysis.

While the evaluation framework employed careful statistical evaluation to discern performance differences, the marked discrepancy in trial counts between classical models and the QSVM exposes a significant limitation. The classical SVMs were assessed over 100 trials, yielding a definitive characterisation of their performance variability. In contrast, the QSVM was evaluated over only 3 trials due to its significantly higher computational demands. This imbalance constrains confidence in fully capturing the QSVM's performance distribution and may inadvertently bias the comparative analysis. Future studies should aim to increase the number of QSVM trials across a wider range of primes to provide a more equitable assessment of their true capabilities. The synthetic nature of the DLP-derived datasets also introduces critical challenges. Although these datasets generate a dataset split equally between classes for large primes, the random data generation process sometimes produces skewed class distributions for smaller primes where the secret s referred to in Appendix B has significantly less possible values. Such imbalances can lead to artificially high accuracies for the more simple, classical models even when these results do not reflect the model's general capability. Although extreme cases of class imbalance were discarded in this methodology, the residual anomalies of this type call into question the reliability of the performance metrics and highlight the need for more sophisticated data generation or augmentation strategies. Incorporating real-world cryptographic datasets in future work may help to mitigate these issues and enhance the generalisability of the findings of this research.

Model Comparison	Mean Accuracy Diff.	p-value
QSVM vs. Linear	17.05%	< 0.00005
QSVM vs. Polynomial	14.41%	< 0.0005
QSVM vs. RBF	8.65%	< 0.05

TABLE I: Two-sample t-test results comparing QSVM to each classical SVM kernel.

Figure 5 further illustrates each model's performance, plotting mean accuracy for datasets of increasing complexity as prime value increases, overlaid with a 4-parameter logistic fit outlined in Appendix E. Although this logistic function provides a convenient way to depict overall trends rather than a physically derived law, it highlights how QSVM maintains a higher accuracy curve relative to classical kernels. Polynomial and RBF SVMs start with reasonable performance on smaller primes but ultimately show much lower performance under more complex conditions. A residual analysis is also performed in Appendix E where one can justifiably attribute the mild heteroscedasticity observed in some residual plots largely to the lack of physical knowledge of the true form of the accuracy trends.

As demonstrated in Figure 4 and Table I, the QSVM significantly outperforms classical analogues in this task, capturing complex non-linear correlations that classical kernels overlook. Nevertheless, the overall performance of the QSVM still exhibited an exponential decline in accuracy as dataset complexity increased, as illustrated in Figure 5. This decline is not attributed to quantum errors (not simulated in this research focussing on the application of quantum algorithms running on classical hardware) but to limitations inherent in the current model architecture, such as fixed circuit depth and limited qubit encoding capacity. Theoretically, enhancing these as-

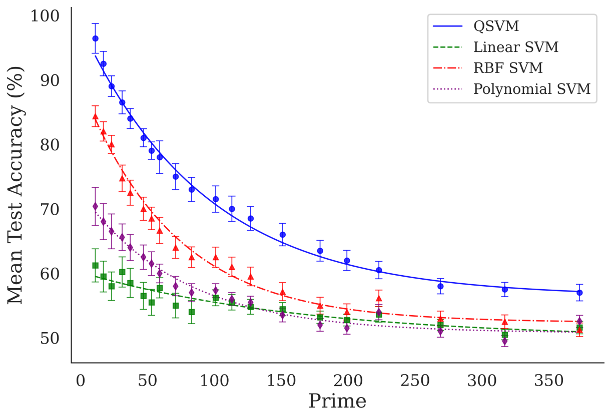
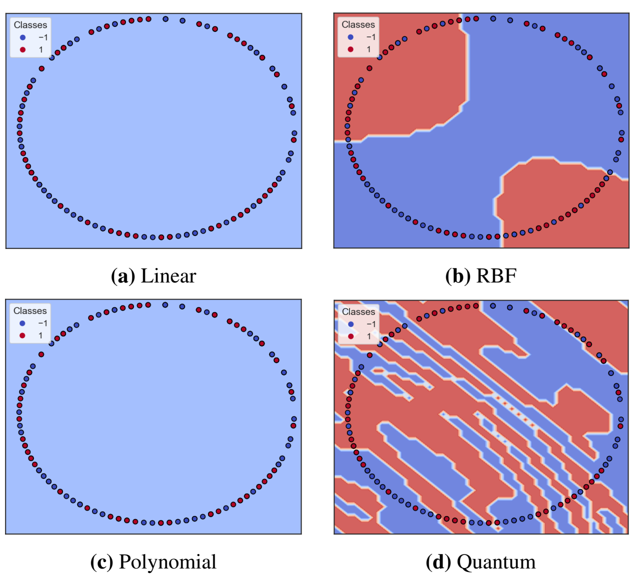


FIG. 5: Accuracy as a function of prime size (used to tune dataset complexity) for each model, overlaid with a 4-parameter logistic fit.

pects by increasing the circuit depth or scaling to more qubits could help maintain or improve QSVM performance on more challenging classification tasks.

Differences in classification behaviour become evident when comparing decision boundaries. Figure 6 illustrates typical boundaries for each kernel type using a dataset derived from $p = 101$ as an example.



(a) Linear **(b)** RBF **(c)** Polynomial **(d)** Quantum

FIG. 6: Comparison of decision boundaries for each kernel on a dataset with $p = 101$. The boundary represents where the model switches its prediction between classes, with red indicating class $+1$ and blue indicating class -1. Correct classes shown by colour of each datapoint.

Figure 6 highlights the significant improvement in expressivity of the decision boundary possible by implementing a quantum kernel. Classical linear and polynomial kernels were incapable of any structure recognition, labelling the entire test set as class: -1; likely due to the training set slightly favouring this class in this case. Whilst the RBF kernel attempts to label some sections of slightly higher density of each class, the quantum-enhanced model exhibits significantly more complex decision making as well as clearly identifying smaller regions of class density throughout the space. The disparity between model expressivity becomes especially apparent as prime size increases, where classical methods tend toward random guessing. For further details and further examples of decision boundaries and confusion matrices, see Appendix E.

To investigate the role of circuit depth the QSVM's performance was evaluated across a range of depths for three representative primes. As shown in Figure 7, datasets derived from simpler data (e.g., $p = 59$) are well-modelled by relatively shallow circuits. In fact, the quadratic fit for $p = 59$ indicates that an optimal circuit depth of approximately $d_{optim} \approx 3$ is reached, with evidence of underfitting at lower depths and overfitting at higher depths. Extrapolating from the quadratic fits for the larger primes, as described in Appendix D, it appears that deeper circuits might be beneficial for capturing the increased complexity of the data. Roughly, the optimal circuit depths for the datasets derived from $p = 101$ and $p = 373$ could be estimated as $d_{optim} \approx 5$ and $d_{optim} \approx 16$ (or at least »5), respectively, based on the fitted quadratics. It is important to note, however, that due to computational limitations only a single run per circuit depth was conducted leading to a lack of statistical error analysis. Therefore, these preliminary results are intended to guide future research by demonstrating that optimal quantum circuit depth for models of this architecture may increase with model complexity rather than serve as a comprehensive analysis.

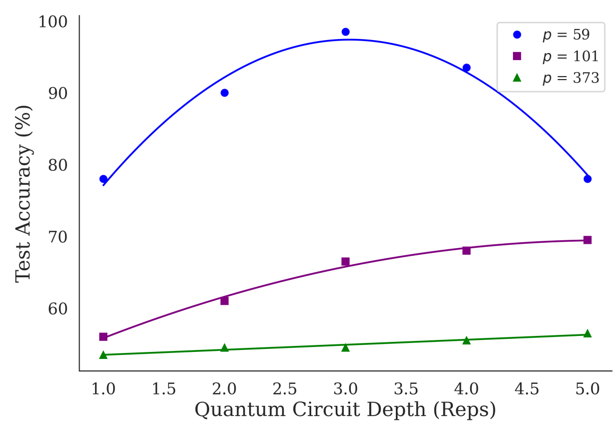


FIG. 7: QSVM accuracy as a function of circuit depth for three different dataset complexities controlled by prime size.

The investigation into circuit depth optimisation highlights the delicate balance between increased expressivity and the risk of overfitting. It is important to emphasise that the estimates for optimal circuit depth are rough approximations, indicating a trend that optimal circuit depth may scale with dataset complexity rather than representing verified optima. Moreover, while deeper circuits are theoretically beneficial for capturing richer feature representations, they also pose practical challenges on real quantum hardware. Increased circuit depth escalates susceptibility to noise and decoherence. With powerful error correction and mitigation strategies, these challenges may be overcome to realise the advantages of deeper circuits in practice. [11]

The application of quantum kernel methods to cryptographic classification tasks indicates a potential shift in understanding of the limitations of machine learning with implications that extend far beyond the realm of cryptography. For instance, in genomics, classical algorithms often struggle with the high-dimensional, non-linear interactions inherent in sequencing data, impeding efforts in personalised medicine [8]. By mapping genomic data into a high-dimensional Hilbert space, quantum kernels could more efficiently identify rare mutations or subtle gene expression patterns, potentially accelerating drug discovery and tailored treatment strategies.

In materials science, the search for novel compounds, such as new superconductors or catalysts, requires the analysis of complex, quantum-level descriptors that classical models cannot fully exploit [12]. Quantum-enhanced methods offer the promise of better feature representations by leveraging entanglement and interference, thus providing more accurate classifications of material properties.

Similarly, in finance, high-frequency trading data and market anomaly detection suffer from non-linear and dynamic interactions that are difficult for classical kernels to capture [9]. Quantum kernel methods may help to reveal hidden market structures, offering improved predictive models.

Climate modelling, another area where classical methodologies struggle due to the volume and complexity of data, stands to benefit substantially. Current climate prediction models require days of processing on supercomputers; quantum-enhanced machine learning could, in theory, reduce training times dramatically by efficiently evaluating vast datasets [10].

Looking ahead, research must tackle two main challenges. First, classical simulation can be improved by approximate methods such as tensor network contractions and variational circuit approximations [5, 13], mitigating the cost of emulating deeper circuits. These methods clarify trade-offs between expressivity and overfitting (see Appendix E). Second, running on real quantum hardware demands significant advances in error mitigation and error correction to combat noise and decoherence. Moreover, hybrid quantum-classical strategies, where quantum components handle high-dimensional embeddings and classical solvers perform optimisation, could scale to real-time tasks across genomics, materials science, finance, and climate modelling. Such progress is key to realising the full potential of quantum-enhanced machine learning in practice.

IV. CONCLUSION

In conclusion, this study introduced a generalised Quantum Support Vector Machine (QSVM) that employs a fidelity-based quantum kernel to map classical data into a high-dimensional Hilbert space, thereby capturing non-linear correlations beyond the reach of classical kernels. In the fixed-depth phase, the QSVM demonstrated statistically significant improvements over classical SVMs in the assigned cryptographic classification task; achieving a mean accuracy improvement of 17.05% over the linear SVM, 14.41% over the polynomial SVM, and 8.65% over the RBF SVM. The subsequent circuit depth optimisation phase provided preliminary evidence that optimal circuit depth increases with dataset complexity. These findings underscore the importance of error mitigation strategies in order to benefit from quantum machine learning techniques when working on large-scale problems requiring real quantum hardware. Looking forward, the integration of quantum kernel methods into hybrid quantum-classical architectures is poised to address real-world applications where classical models struggle. Domains such as genomics, advanced materials, finance, and climate modelling that involve high-dimensional, non-linear data that could benefit from the faster training times and richer feature embeddings afforded by quantum techniques. As quantum hardware evolves with improved error correction, the methodologies and architectures developed in this work provide a compelling foundation for translating theoretical advantages into practical machine learning solutions.

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SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

This study explores how principles of quantum physics (a theory that describes the behaviour of extremely small particles) can enhance a special type of computer program that is used to sort information into groups. Normally, these programs work by drawing a line or boundary to separate different types of information into categories. This can be useful for simple tasks such as separating spam emails from important ones in your inbox. However, when using these programs for more challenging tasks, where differences between the categories are difficult to spot (for example, distinguishing early signs of cancer in medical imaging), this approach may fail.

This research aims to build a new type of program to handle these more challenging tasks by taking advantage of the unusual way quantum particles behave. Firstly, small particles can exist in more than one place at the same time. Also, pairs of these particles can be invisibly connected even when far apart, allowing them to share information in unique ways. These properties allow a new program called a “quantum support vector machine” (QSVM) to look at information in a new, more intelligent way, where hidden patterns can be easier to spot.

This research built and tested the QSVM on tasks that require complex pattern spotting to crack codes and found that the new program was better at sorting new information into the correct groups than older methods. Further investigation also suggested that adding more “quantum behaviour” to the program could help improve accuracy, but only up to a certain point before it started to work less well.

In general, this work suggests that the use of these new quantum-enhanced programs could open up new possibilities for solving complex problems in many areas, such as medicine, finance, and climate research, where using information to make predictions is very important.

STATEMENT ON THE USE OF GENERATIVE AI TOOLS

Generative AI tools were used on occasion to help with minor language editing and phrasing throughout this work. Their use was limited to refining the presentation of the text and does not extend to the generation of technical content, data analysis, or original research ideas.

Appendix A: Classical Support Vector Machines

1. Overview of SVM Theory and Geometry

Support Vector Machines (SVMs) are powerful classifiers that aim to separate data into two classes by finding an optimal hyperplane in a (potentially) high-dimensional feature space. Consider a training dataset

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$. In the simplest linearly separable case, the primal optimisation problem is

$$\begin{aligned} \text{minimise:} \quad & \frac{1}{2}\|\mathbf{w}\|^2 \\ \text{subject to:} \quad & y_i\big(\mathbf{w}\cdot\mathbf{x}_i + b\big) \;\geq\; 1, \quad \forall\, i = 1, \ldots, N \end{aligned}$$

Here, $\mathbf{w}$ is the normal vector to the separating hyperplane, and b is the bias term. The objective $\frac{1}{2}\|\mathbf{w}\|^2$ is equivalent to maximising the margin (distance) between the hyperplane and the closest training samples, i.e. the support vectors.

Introducing Lagrange multipliers $\alpha_i \geq 0$ for each constraint, one can eliminate $\mathbf{w}$ and b to obtain the dual problem:

$$\begin{aligned} \text{maximise:} \quad & \sum_{i=1}^{N}\alpha_i \;-\; \frac{1}{2}\sum_{i,j=1}^{N}\alpha_i\,\alpha_j\,y_i\,y_j\big(\mathbf{x}_i\cdot\mathbf{x}_j\big) \\ \text{subject to:} \quad & \sum_{i=1}^{N}\alpha_i\,y_i \;=\; 0 \quad \alpha_i \;\geq\; 0 \end{aligned}$$

Many α_i become zero at the optimum, and only the data with non-zero α_i (the support vectors) explicitly shape the classification boundary. The final decision function for an input $\mathbf{x}$ is

$$f(\mathbf{x}) = \operatorname{sign}\Big(\sum_{i=1}^{N}\alpha_i\,y_i\,(\mathbf{x}_i\cdot\mathbf{x}) \;+\; b\Big)$$

2. Kernel Trick for Non-linear Separations

Real-world data are often not linearly separable. To handle non-linearities, SVMs replace the dot product $(\mathbf{x}_i \cdot \mathbf{x}_j)$ with a kernel $K(\mathbf{x}_i, \mathbf{x}_j)$. By definition,

$$K(\mathbf{x}, \mathbf{x}') = \big\langle \phi(\mathbf{x}),\, \phi(\mathbf{x}') \big\rangle$$

where ϕ is a (possibly high-dimensional) feature map. Common examples include:

- **Linear:** $K_{\text{lin}}(\mathbf{x}, \mathbf{x}') = \mathbf{x}\cdot\mathbf{x}'$
- **Polynomial:** $K_{\text{poly}}(\mathbf{x}, \mathbf{x}') = \big(\mathbf{x}\cdot\mathbf{x}' + c\big)^d$
- **RBF (Gaussian):** $K_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\big[-\gamma\,\|\mathbf{x}-\mathbf{x}'\|^2\big]$

Replacing $\mathbf{x}_i \cdot \mathbf{x}_j$ with $K(\mathbf{x}_i, \mathbf{x}_j)$ in the dual formulation enables non-linear decision boundaries to emerge in the original $\mathbf{x}$-space.

Appendix B: Dataset Construction

1. Prime-based Modular Data Generation

To generate cryptographic-style data, this research exploits the Discrete Logarithm Problem (DLP). Letting p be a prime, the multiplicative group $\mathbb{Z}_p^*$ then contains $\{1, 2, \ldots, p-1\}$. A generator g (also called a "primitive root") of this group satisfies:

$$g^{\,k} \bmod p \quad \text{yields every residue in } \{1, \ldots, p-1\} \text{ as } k \text{ goes from 1 to } p-1$$

To find g, one checks that

$$g^{\,(p-1)/q_i} \not\equiv 1 \pmod{p} \quad \text{for every prime factor } q_i \text{ of } (p-1)$$

2. Residue Sampling and Labelling

Data is synethesised for the classification task as follows:

1. **Exponent Sampling:** Sample an integer y uniformly in $\{1, \ldots, p-1\}$.
2. **Residue Computation:** Compute $x \equiv g^y \bmod p$.
3. **Label Generation (Secret Shift):** Choose a secret integer $s \in \{1, \ldots, p-1\}$. Assign the class label (that defines the target variable for each model to predict).

$$\ell = \begin{cases} +1, & \text{if } s \;\le\; y < s + \frac{p-1}{2} \pmod{p-1}, \\ -1, & \text{otherwise} \end{cases}$$

Hence each data point $(x,\, \ell)$ depends on exponent y, but only x (and the label) are exposed to the model during testing, capturing a cryptographically relevant, non-linear relationship.

3. Geometric Feature Map to $(\sin, \cos)$

The integer x is mapped onto the unit circle for 2D representation:

$$x_{\sin} = \sin\Bigl(\frac{2\pi\, x}{p}\Bigr), \quad x_{\cos} = \cos\Bigl(\frac{2\pi\, x}{p}\Bigr)$$

Thus each sample is

$$\bigl(x_{\sin},\, x_{\cos}\bigr) \;\in\; \mathbb{R}^2$$

labelled by $\ell \in \{\,-1,\, +1\}$. Small primes yield simpler class boundaries; larger primes produce increasingly intricate patterns due to the size of the multiplicative group, $\mathbb{Z}_p^*$, increasing with p.

Appendix C: Quantum Feature Map and Kernel

1. Quantum States and the Born Rule

In quantum computing, an n-qubit system is described by a state vector $|\Psi\rangle$ in a 2^n-dimensional Hilbert space $\mathcal{H}_{2^n}$. The probability of measuring a particular basis state $|m\rangle$ is

$$\Pr(\text{outcome} = m) = \bigl|\langle m \mid \Psi\rangle\bigr|^2$$

as prescribed by the Born rule. Instead of performing standard projective measurements for classification, the fidelity between encoded states is exploited to build the kernel function.

2. Parametrised Feature Maps

A parametrised quantum circuit (PQC) $U_\phi(\mathbf{x})$ serves as a tool to encode classical data $\mathbf{x}$ into a quantum state. In this process, the circuit transforms a simple initial state (usually the all-zeros state) into a quantum state in a higher dimensional Hilbert Space.

$$|\phi(\mathbf{x})\rangle = U_\phi(\mathbf{x})\,|0\rangle^{\otimes d}$$

In this experimental setup, consider a feature space of dimension $d = 2$ where the features are given by $(x_{\sin},\, x_{\cos})$. Here, each classical feature is directly mapped to an individual qubit, ensuring that the data is distributed across the quantum register.

As detailed in the methods section, a specific unitary operator is used to map the input features into a highly entangled Hilbert space. This unitary is constructed by repeatedly applying layers of gates, each layer encoding the data and introducing entanglement. The operator is given by:

$$U_\Phi(\mathbf{x}) = \prod_{k=1}^{\mathrm{Reps}} \Bigg[\underbrace{\prod_{i=1}^{D} R_Z(x_i)}_{\text{single-qubit } Z\text{-rotations}} \quad \underbrace{\prod_{i<j}^{D} R_{YY}(x_i, x_j)}_{\text{two-qubit entangling}} \Bigg] \tag{C1}$$

This expression indicates that for each of the Reps (repetitions, or depth layers) of the circuit, the following operations are performed:

- **Single-Qubit R_Z Rotations:** For each qubit i (with $i = 1, \ldots, D$, where D is the number of features), a rotation about the Z-axis is applied. The gate $R_Z(x_i)$ is defined as

  $$R_Z(x_i) = e^{-i\,x_i\,Z/2}$$

  meaning that the rotation angle is directly related to the corresponding input feature x_i. These rotations encode the classical data into the phase of each qubit.

- **Two-Qubit R_{YY} Entangling Rotations:** For every pair of distinct qubits i and j (with $i < j$), an entangling operation is applied using the R_{YY} gate. This gate performs a rotation about the $Y \otimes Y$ axis and is generally defined as

  $$R_{YY}(x_i, x_j) = e^{-i\,\theta_{ij}\,(Y \otimes Y)/2}$$

  where the angle θ_{ij} is typically a function of the pair of features (x_i, x_j). This gate not only encodes the relationship between features into the quantum state but also introduces entanglement between the qubits—an essential resource for quantum machine learning.

In the overall construction:

- Reps represents the number of layers or repeating blocks in the circuit. Increasing the number of repetitions enhances the expressivity of the embedding, allowing the circuit to capture more complex correlations in the input data.
- D is the feature dimension, which in this case is equal to the number of qubits.
- The single-qubit rotations $R_Z(x_i)$ provide a straightforward mechanism to encode each individual feature.
- The two-qubit rotations $R_{YY}(x_i, x_j)$ serve to entangle the qubits, thereby correlating the features in a non-classical manner.

This construction is particularly powerful because it forms a fairly generalised quantum embedding operator. By appropriately choosing the number of repetitions and the specific Pauli rotations, one can tailor the feature map to suit the problem at hand, potentially improving the performance of quantum kernel methods in classification tasks.

The exact form of the quantum circuit implementing $U_\Phi(\mathbf{x})$ is depicted in Figure 8. This figure shows the fully decomposed circuit where each gate's operation is clearly indicated. A corresponding key, provided in Figure 9, details the individual gate types, such as the R_Z rotations, basis change rotations (which may be used to convert the Y basis to the computational basis for the entangling gates), and the CNOT gates that appear in the implementation of the R_{YY} rotation.

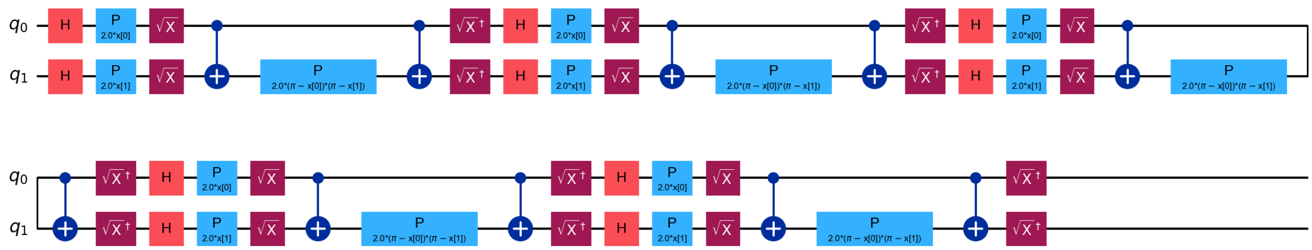


FIG. 8: Unitary quantum circuit, $U_\Phi(\mathbf{x})$, used for feature mapping during fixed depth analysis with depth (reps) = 5.

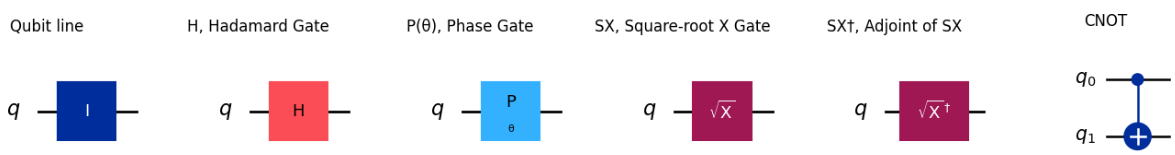


FIG. 9: A Key detailing each quantum gate type used in the feature map circuit.

To illustrate how the circuit realises the unitary $U_\Phi(\mathbf{x})$ in practice, consider the following detailed description of a specific sequence of gates in one of the layers:

1. **Hadamard Gate (H):** The circuit begins with a Hadamard gate applied to a qubit, which transforms the basis state $|0\rangle$ into an equal superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}}\Big(|0\rangle + |1\rangle\Big)$$

This superposition is critical as it prepares the qubit so that subsequent phase rotations can create interference patterns that encode the data.

2. **Phase Gate $P_{2.0\cdot x[1]}$:** Following the Hadamard gate, a phase gate is applied with a rotation angle proportional to the feature $x[1]$. The phase gate is defined by

$$P_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

Setting $\phi = 2x[1]$ yields

$$P_{2.0\cdot x[1]} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\,2x[1]} \end{pmatrix}$$

This gate rotates the state about the Z-axis, encoding the classical feature $x[1]$ into the relative phase between the qubit's basis states.

3. **Square-root X Gate ($\sqrt{X}$):** Next, the $\sqrt{X}$ gate is applied. It is defined as

$$\sqrt{X} = e^{-i\frac{\pi}{4}X} = \cos\left(\frac{\pi}{4}\right) I - i\sin\left(\frac{\pi}{4}\right) X$$

which performs a rotation about the X-axis by $\pi/2$. This operation changes the qubit's state into a new basis that is particularly suited for the subsequent entangling step.

4. **Entangling Operation (CNOT Gate):** An entangling operation is then applied between two qubits using a controlled-NOT (CNOT) gate. In the circuit diagram, this is represented by a line joining the qubits with a plus sign, indicating the control-target relationship. The CNOT gate acts on two qubits: if the control qubit is in the state $|1\rangle$, it flips the state of the target qubit; if the control qubit is in the state $|0\rangle$, the target qubit remains unchanged. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the matrix representation of the CNOT gate is

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

For example, if the control qubit is in a superposition state $\alpha|0\rangle + \beta|1\rangle$ and the target qubit is initialised to $|0\rangle$, then the joint state is

$$(\alpha|0\rangle + \beta|1\rangle)|0\rangle$$

After applying the CNOT gate (with the first qubit as control), the resulting state is

$$\alpha|0\rangle|0\rangle + \beta|1\rangle|1\rangle$$

which is an entangled state provided α and β are both non-zero. This entanglement is essential for capturing the interdependencies between features.

5. **Two-Qubit Phase Gate $P_{2.0\cdot(\pi-x[0])\cdot(\pi-x[1])}$:** Finally, a two-qubit phase gate is applied. This gate, analogous to the single-qubit phase gate but acting jointly on two qubits, has a rotation angle given by $2.0 \cdot (\pi - x[0]) \cdot (\pi - x[1])$. Mathematically, this can be written as

$$P_{2.0\cdot(\pi-x[0])\cdot(\pi-x[1])} = \exp\Big(-i\,(\pi - x[0])(\pi - x[1])\Big)$$

This gate further entangles the qubits by introducing a phase that is a non-linear function of the features $x[0]$ and $x[1]$. Its role mirrors that of the R_{YY} gate in the general construction:

$$R_{YY}(x_i, x_j) = e^{-i\,\theta_{ij}\,(Y\otimes Y)/2}$$

with the specific choice $\theta_{ij} = 2.0 \cdot (\pi - x[0]) \cdot (\pi - x[1])$.

Together, these operations are mathematically equivalent to applying single-qubit rotations that encode the data followed by two-qubit R_{YY} entangling rotations, thereby reproducing the desired unitary transformation $U_\Phi(\mathbf{x})$. Overall, the parametrised feature map $U_\Phi(\mathbf{x})$ transforms the classical input vector $\mathbf{x}$ into a quantum state that reflects both individual feature contributions and their mutual correlations, thereby laying the groundwork for improved performance in quantum kernel methods via a reasonably generalised framework.

3. Fidelity-based Quantum Kernel

Given two data points $\mathbf{x}$ and $\mathbf{y}$:

$$|\phi(\mathbf{x})\rangle = U_\phi(\mathbf{x})|0\rangle^{\otimes d}, \quad |\phi(\mathbf{y})\rangle = U_\phi(\mathbf{y})|0\rangle^{\otimes d}$$

The quantum kernel is defined as

$$K_Q(\mathbf{x}, \mathbf{y}) = \Big|\,\langle\phi(\mathbf{x})\,|\,\phi(\mathbf{y})\rangle\Big|^2$$

The use of fidelity as a kernel measure is motivated by its role as a natural metric in Hilbert space, quantifying the geometric overlap between quantum states. This intrinsic property ensures that the resulting quantum kernel is symmetric and positive semi-definite, satisfying Mercer's conditions. Substituting K_Q into an SVM yields a QSVM whose decision boundaries can reflect the higher-order correlations discernible as a result of superposition and entanglement.

Appendix D: Circuit-Depth Optimisation

1. Motivation

Increasing the Reps (number of repeated layers in the feature map) generally augments the expressive power of the parametrised quantum circuit (PQC) by incorporating additional single-qubit rotations and entangling operations. This increased expressivity, however, is accompanied by two significant risks:

- **Overfitting in Noise-Free Simulations:** In a simulation environment free from hardware noise, an overly flexible feature map may fit the training data too closely, capturing spurious patterns rather than the underlying structure. This phenomenon reduces generalisation performance on unseen data.
- **Accumulation of Errors on NISQ Devices:** On current noisy intermediate-scale quantum (NISQ) devices, every additional gate increases the probability of errors through decoherence and imperfect gate operations. Thus, while deeper circuits can in principle better capture complex data correlations, they may also yield diminished performance due to increased noise.

Additionally, deeper circuits may require greater computational resources, both for simulation and for processing larger kernel matrices.

2. Algorithmic Depth-Sweep

A systematic approach was adopted to balance the benefits and drawbacks of increasing circuit depth. The algorithmic depth-sweep procedure is outlined as follows:

1. **Initial Depth:** Begin with $\mathrm{Reps} = 1$, corresponding to the simplest version of the quantum feature map.
2. **Compute Kernel:** For each depth d, the Gram matrices K_{train} and K_{test} are computed for the training and test datasets, respectively, using a fidelity-based quantum kernel derived from the PQC.
3. **Train & Evaluate:** The quantum support vector machine (QSVM) is trained using K_{train} and subsequently evaluated on the test set using K_{test}. Both training and test accuracies are recorded.
4. **Increment Depth:** The depth is incremented ($\mathrm{Reps} \to \mathrm{Reps} + 1$) and the process is repeated. The sweep continues until the test accuracy plateaus or begins to decline.

This iterative procedure mirrors hyperparameter tuning in classical machine learning (e.g., varying the polynomial degree or adjusting the RBF kernel's γ parameter), allowing one to identify a circuit depth that optimally balances expressivity and generalisation.

3. Fitted Quadratics for Each Prime: Details and Analysis

To quantitatively assess the relationship between circuit depth and classification accuracy, the test accuracies recorded across varying depths were fitted with quadratic functions. Although the quadratic form is not derived from first-principles physical laws, it provides a convenient empirical model to approximate the observed trend in accuracy as a function of depth.

For each prime p under investigation (e.g., $p = 59$, $p = 101$, and $p = 373$), the following steps were performed:

1. **Data Collection:** Using the procedure described above, for each depth d in the chosen sweep range (for instance, $d = 1, 2, \ldots, 10$ for $p = 101$), both training and test accuracies were computed. In the code implementation (see `depth_optimisation.py`), a stopping criterion was incorporated: if the test accuracy declined for three consecutive depths, the sweep was terminated early.
2. **Quadratic Fitting:** The measured test accuracies $A(d)$ were then fitted to a quadratic model of the form:
$$A(d) = a\,d^2 + b\,d + c$$
where the coefficients a, b, and c were determined via least-squares regression.
3. **Optimal Depth Estimation:** The vertex of the quadratic, given by
$$d_{\mathrm{opt}} = -\frac{b}{2a}$$
serves as an estimate of the optimal circuit depth. For simpler datasets (e.g., those generated using smaller primes), the fitted quadratic typically peaks at lower depths (around $d \approx 3$). Conversely, for more complex datasets (larger primes), the optimal depth shifts higher. Preliminary results suggest $d_{\mathrm{opt}} \approx 5$ for $p = 101$ and potentially $d_{\mathrm{opt}} \gg 5$ (e.g., $d_{\mathrm{opt}} \approx 16$) for $p = 373$.
4. **Analysis of Overfitting Trends:** The quadratic fit not only provided an estimate for d_{opt} but also offered insight into the overfitting behaviour. At depths exceeding the optimal value, the decline in test accuracy (despite potential improvements in training accuracy) was interpreted as a symptom of overfitting in the noise-free simulation.

In summary, this phase of the methodology implements a practical and systematic method to tune the complexity of the quantum feature map (for a fixed number of features/qubits). The use of quadratic fits provides an intuitive estimate of the optimal depth, balancing expressivity against the risk of overfitting, and sets the stage for future work incorporating noise and error mitigation strategies on real quantum hardware.

Appendix E: Extended Experimental Results

1. Full Accuracy Tables

Table II compiles the mean test accuracies (with standard errors in parentheses) for each model across a range of prime values. These results are averaged over multiple runs with different seeds and train/test splits. Notice that for the smallest prime ($p = 3$) all models achieve perfect accuracy due to the simplicity of the generated dataset. However, as p increases and the data become more complex, the QSVM consistently outperforms the classical models.

TABLE II: Mean Test Accuracies (%) with Standard Errors for Different Primes.

p	QSVM (SE)	Linear SVM (SE)	RBF SVM (SE)	Poly SVM (SE)
3	100.0 (0.00)	100.0 (0.00)	100.0 (0.00)	100.0 (0.00)
11	96.4 (2.31)	61.22 (2.57)	84.35 (1.61)	70.37 (2.94)
17	92.5 (1.90)	59.50 (2.40)	82.00 (1.50)	68.00 (2.80)
23	89.0 (1.60)	58.00 (2.20)	80.00 (1.40)	66.50 (2.70)
31	86.5 (1.78)	60.21 (2.35)	74.72 (2.06)	65.59 (2.07)
37	84.0 (1.55)	58.50 (2.25)	72.50 (1.95)	64.00 (2.00)
47	81.0 (1.40)	56.50 (2.10)	70.00 (1.80)	62.50 (1.90)
53	79.0 (1.35)	55.50 (2.00)	68.50 (1.75)	61.50 (1.85)
59	78.0 (2.52)	57.75 (1.55)	66.64 (2.02)	59.99 (1.45)
71	75.0 (2.00)	55.00 (1.90)	64.00 (1.70)	58.00 (1.50)
83	73.0 (1.85)	54.00 (1.80)	62.50 (1.60)	57.00 (1.40)
101	71.5 (2.04)	56.21 (1.24)	62.50 (1.57)	57.33 (1.05)
113	70.0 (1.95)	55.50 (1.20)	61.00 (1.50)	56.00 (1.00)
127	68.5 (1.85)	54.80 (1.15)	59.50 (1.45)	55.50 (0.95)
151	66.0 (1.75)	54.45 (1.02)	57.15 (1.40)	53.53 (1.08)
179	63.5 (1.60)	53.20 (0.95)	55.00 (1.30)	52.00 (1.00)
199	62.0 (1.55)	52.75 (0.90)	54.00 (1.25)	51.50 (0.95)
223	60.5 (1.34)	53.63 (1.20)	56.14 (1.26)	53.98 (1.18)
269	58.0 (1.20)	52.00 (0.85)	53.00 (1.15)	51.00 (0.90)
317	57.5 (1.10)	50.50 (0.80)	52.50 (1.05)	49.50 (0.85)
373	57.0 (1.28)	51.60 (0.97)	51.25 (1.06)	52.55 (0.93)

2. Statistical Significance Tests

To assess the observed performance differences, a standard two-sample t-test was performed. Let

$$\{A_{\mathrm{Q}}^{(r)}\}_{r=1}^{R} \quad \text{and} \quad \{A_{\mathrm{C}}^{(r)}\}_{r=1}^{R}$$

be the sets of accuracies obtained from multiple runs (with different seeds and train/test splits) for the QSVM and a given classical model, respectively. The null hypothesis $H_0 : \mathbb{E}[A_{\mathrm{Q}}] = \mathbb{E}[A_{\mathrm{C}}]$ was tested against the alternative hypothesis that the QSVM mean is higher. The results of these tests are summarised in Table III.

TABLE III: Comparison of QSVM with Classical SVM Models.

Comparison	Mean Diff. (%)	t-statistic	p-value	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.001$
QSVM vs. Linear SVM	17.05	4.71	3.33×10^{-5}	True	True	True
QSVM vs. RBF SVM	8.65	2.17	0.0356	True	False	False
QSVM vs. Poly SVM	14.41	3.86	0.000413	True	True	True

3. Residual Analysis and Logistic Fits

To understand how the classification accuracy scales with increasing prime value p, the mean accuracies are plotted against $\log(p)$. The resulting trend often appears sigmoidal, reflecting a gradual degradation in performance as the dataset complexity increases. To capture this behaviour, the accuracy data was fitted using a 4-parameter logistic function of the form

$$A(p) = a + \frac{b - a}{1 + \left(\frac{p}{p_0}\right)^{-\Delta}}$$

where:

- a represents the lower asymptote (the minimum achievable accuracy),
- b represents the upper asymptote (the maximum accuracy in simpler cases),
- p_0 is the inflection point, and
- Δ controls the slope of the transition.

It is important to note that this logistic function is an empirical model and is not physically motivated. It is used primarily as a convenient tool for visualisation and trend estimation. Therefore, while the fitted parameters a, b, p_0, and Δ provide a useful description of the overall trend, they should not be over-interpreted in a physical sense.

Residuals, defined as

$$\text{Residual} = A_{\text{empirical}} - A(p)$$

were analysed to assess the quality of the fit. In the classical models, the residuals exhibit some heteroscedasticity (i.e., a non-uniform spread across the range of p), indicating that the logistic function may not capture all the nuances of the accuracy decay. In contrast, the QSVM residuals exhibit not only a low overall variance but also a pronounced periodic trend as a function of p. This clear periodic behaviour, which is not adequately captured by the logistic fit, suggests that systematic oscillations inherent in the data generation process affect the QSVM's scaling performance. A more detailed analysis of these periodic patterns could yield deeper insights into the interplay between the quantum kernel architecture and the dataset complexity.

Figure 10 illustrates the residual distributions for the logistic fit. The QSVM residuals (blue) show a narrower spread with evident periodic fluctuations, while the classical models exhibit a broader, less uniform distribution.

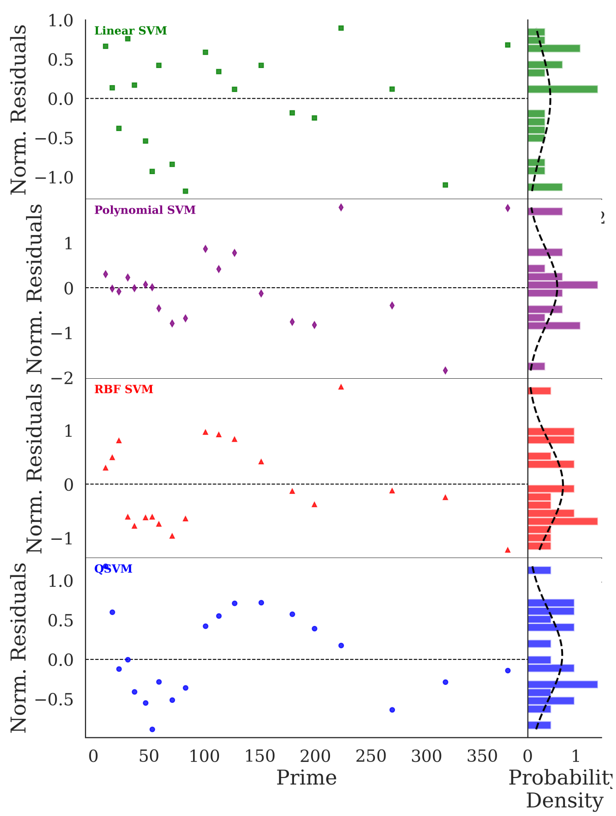


FIG. 10: Residual distributions for the logistic fit to the accuracy data. The QSVM residuals (blue) exhibit a pronounced periodic trend and lower overall variance compared to the classical SVM models.

4. Parameter Fit Analysis

In addition to the logistic fits of the accuracy trends, a detailed parameter estimation was performed for each model. Table IV summarises the goodness of fit of the logistic function for each model including coefficient of determination R^2 and reduced chi-squared. These quantitative measures provide further insight into the certainty of scaling behaviour of each model.

TABLE IV: Comparison of Models: R^2 and Reduced Chi-Squared.

Model	R^2	χ^2_{red}
QSVM	0.99	0.38
Linear SVM	0.86	0.49
Poly SVM	0.98	0.89
RBF SVM	0.98	0.77

The parameter fit analysis reveals significant differences among the models. Notably, the QSVM parameters yield a high R^2 value and low reduced chi-squared, confirming a good fit overall. However, the evident periodic oscillations in the residuals (as discussed in Section E) indicate that the logistic model, while useful for trend estimation, does not fully capture the systematic variations in the QSVM performance. This observation motivates further investigation into alternative fitting models that might better account for the periodicity. An improved understanding of accuracy degradation trends on more complex datasets will be vital for scaling to real-world problems.

5. Example Decision Boundaries and Confusion Matrices for Representative Primes

This subsection provides a graphical illustration of how each model (Linear, Polynomial, RBF, and QSVM) classifies DLP-inspired datasets for three representative primes: $p = 59$, $p = 101$, and $p = 233$. For each prime, this section presents:

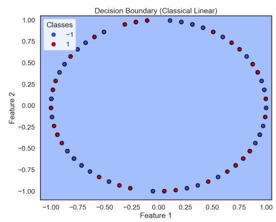
- A decision boundary figure showing how the four models partition $(x_{\sin}, x_{\cos})$-space.
- A corresponding confusion matrix figure illustrating correct (diagonal entries) vs. incorrect (off-diagonal entries) classifications.

By placing the confusion matrices immediately after the decision boundaries for each prime, readers can easily correlate visually misclassified regions with the off-diagonal entries in the confusion matrices. The following pages provide results for $p = 59$, $p = 101$, and $p = 233$, each with a brief caption and then a collective analysis discussing key observations and trends.

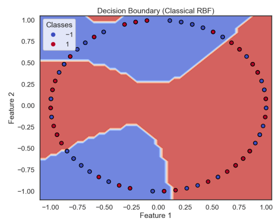
Interpretation

- **Decision Boundaries Interpretation:** Each 2D boundary plot shown in Figures 11,12,13 shows how the classifier partitions the $(x_{\sin}, x_{\cos})$-plane. Regions labeled ± 1 indicate predicted classes. As p increases ($p = 101$ and $p = 233$), classical kernels fail to adapt adequately to highly intricate distributions, sometimes defaulting to constant predictions with essentially no pattern recognition, whereas the QSVM maintains nuanced separations.
- **Confusion Matrices Interpretation:** The diagonal elements represent correct classifications (true positives and true negatives), while off-diagonal cells correspond to misclassifications (false positives and false negatives). When the data complexity increases, classical methods frequently exhibit reduced diagonal occupancy, indicating a higher rate of errors. By contrast, the QSVM tends to retain stronger diagonal elements, consistent with the quantitative accuracy improvements outlined in previous sections.
- **Overall Observations:** Across all primes tested, the QSVM demonstrates an advantage in both boundary delineation and classification accuracy. The entangled quantum feature space captures non-linear relationships that simpler kernels (e.g., linear or polynomial) cannot access. While the classical RBF kernel partially handles non-linearity, it does not match the rich expressivity of the QSVM's fidelity-based kernel on these cryptographic-style datasets.

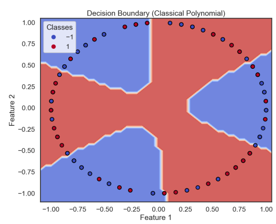
Prime $\boldsymbol{p = 59}$



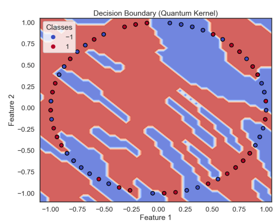
(a) Linear



(b) RBF

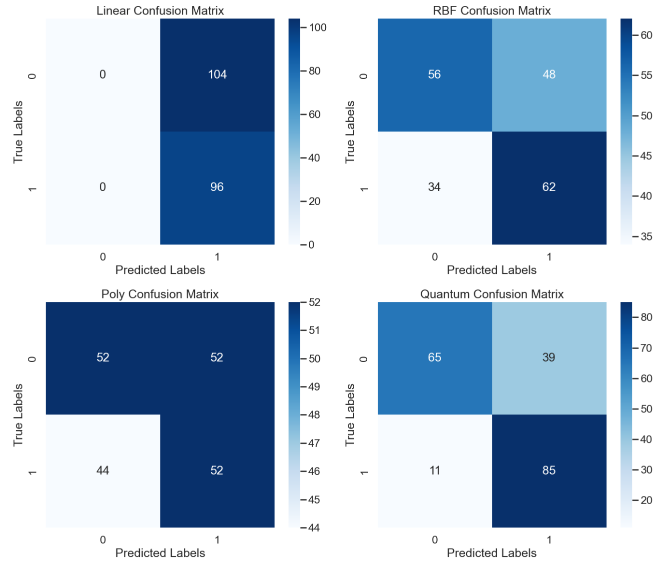


(c) Polynomial



(d) Quantum

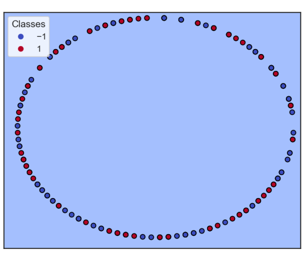
(e) Decision Boundaries for $p = 59$.



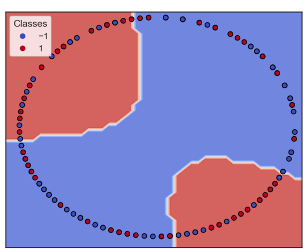
(f) Confusion Matrices for $p = 59$.

FIG. 11: Model comparisons for $p = 59$. The left panel shows how each SVM variant partitions the $(x_{\sin}, x_{\cos})$ plane; the right panel displays the corresponding confusion matrices, where darker diagonal cells indicate higher correct classification.

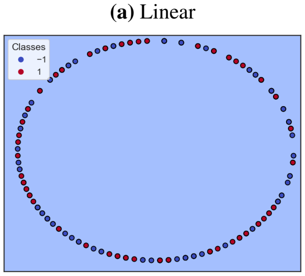
Prime $\boldsymbol{p = 101}$



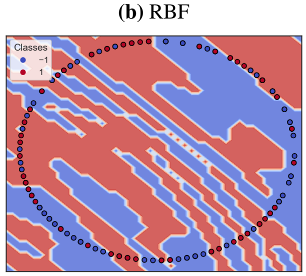
(a) Linear



(b) RBF

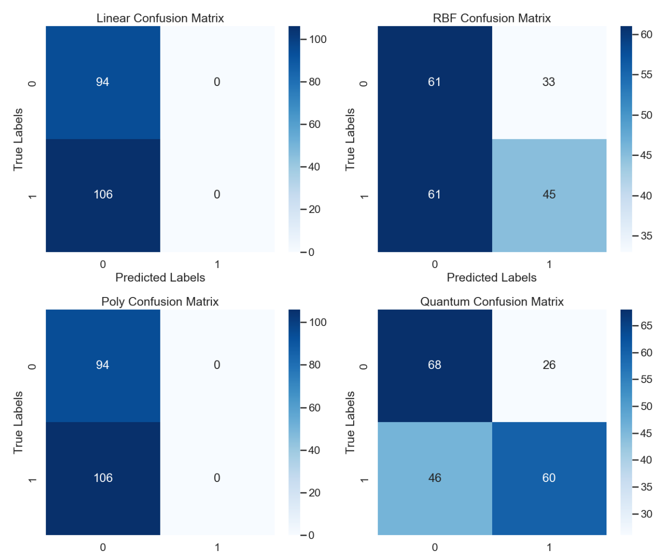


(c) Polynomial



(d) Quantum

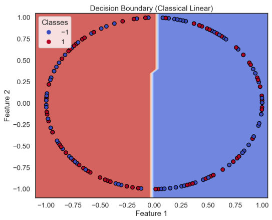
(e) Decision Boundaries for $p = 101$.



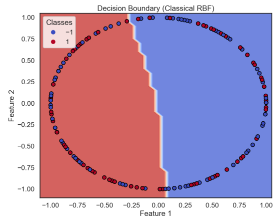
(f) Confusion Matrices for $p = 101$.

FIG. 12: Model comparisons for $p = 101$. The dataset complexity increases with a larger prime, intensifying class overlaps. Classical kernels often misclassify complex boundaries, while the QSVM retains a more precise decision boundary and higher diagonal values in the confusion matrix.

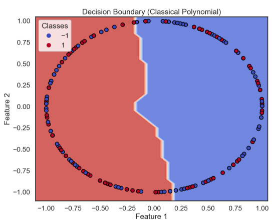
Prime $\boldsymbol{p = 233}$



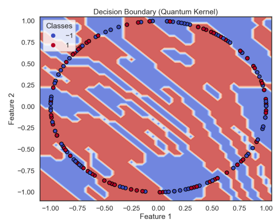
(a) Linear



(b) RBF

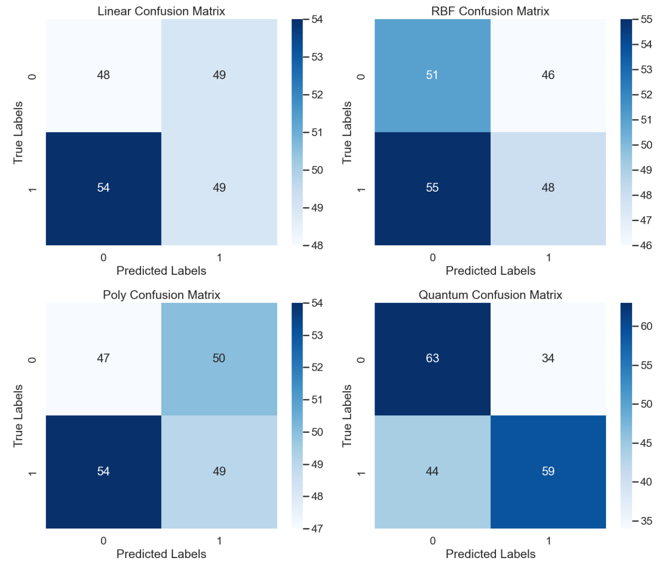


(c) Polynomial



(d) Quantum

(e) Decision Boundaries for $p = 233$.

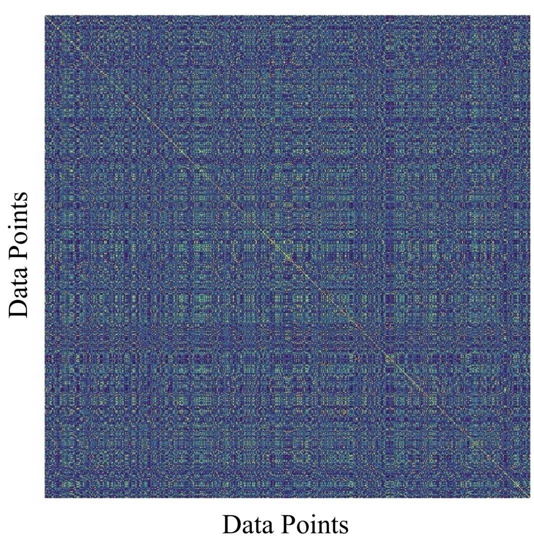


(f) Confusion Matrices for $p = 233$.

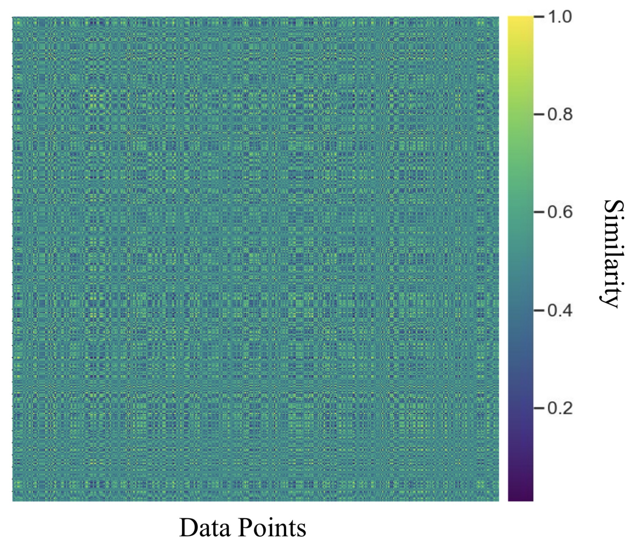
FIG. 13: Model comparisons for $p = 233$. At this larger prime, Linear and Polynomial kernels frequently misclassify large portions of the dataset. While RBF partly mitigates non-linear effects, the QSVM still displays fewer misclassifications overall.

Comparing Linear vs. Quantum Kernel Heatmaps

Although the decision boundaries and confusion matrices clarify the overall performance of each model, it can be equally insightful to examine some examples of the kernel matrices themselves. Figure 14 compares the pairwise similarity (kernel) matrices for the Linear kernel (left) and the Quantum kernel (right) on a subset of an example dataset. Each cell in these matrices represents the similarity between two data points (indexed along the axes), with the colour intensity ranging from 0 (light) to 1 (dark), as indicated by the colour bar.



(a) Quantum Kernel Matrix



(b) Linear Kernel Matrix

FIG. 14: Comparison of linear vs. quantum kernel matrices. Each matrix cell shows the pairwise similarity of two data points, with the colour bar on the right indicating the scale from 0 (minimal similarity) to 1 (maximal overlap). Note that for both maps there is a noticeably bright diagonal as each point is, of course, maximally similar to itself.

Interpretation

- **Linear Kernel:** The mostly bright cells indicate that many data points appear "similar" to one another under a simple dot product in $(x_{\sin}, x_{\cos})$-space. In cryptographic contexts, this can be misleading, causing the model to lump together points from different classes and reducing its ability to carve out intricate decision boundaries. This visualisation suggests that it overestimates similarities between points.

- **Quantum Kernel:** The darker baseline indicates that not every pair of points is treated as strongly overlapping. Instead, genuinely related pairs stand out with higher intensity (closer to 1), indicating the entangled feature map picks up on complex, non-linear structures. This more selective pattern often translates into better class discrimination, as seen in the QSVM's improved confusion matrices and decision boundaries.

- **Overall Significance:** These contrasting heatmaps reveal why the QSVM can outperform simpler kernels: the quantum embedding reshapes the data such that subtle relationships are amplified, rather than indiscriminately assigning high similarities across the board.

Taken together, these visual results confirm that quantum-enhanced kernels provide a consistent advantage for classification, especially as prime size grows and the underlying data distribution becomes more intricate. While classical kernels plateau in their capacity to discriminate complex patterns, the QSVM's entangled feature map leverages higher-dimensional correlations. However, one must balance these benefits against the computational cost of simulating quantum fidelity matrices and plotting large, high-resolution figures for larger primes.

Appendix F: Further Physical Considerations

Although this research primarily conducts noiseless simulations in a statevector environment, real quantum hardware introduces practical constraints that can affect both performance and feasibility. This appendix discusses some of these physical considerations, including noise and error mitigation strategies, the trade-off between entanglement and overfitting, scalability to large datasets, and potential hybrid quantum-classical enhancements.

1. Noise and Error Mitigation in NISQ Devices

Current quantum processors are classified as noisy intermediate-scale quantum (NISQ) devices, where limited qubit counts and imperfect gate operations give rise to decoherence, readout errors, and gate errors. While this experiment's simulations ignore these effects, any real hardware implementation must address them:

- **Short-Depth Ansatz:** Since each additional gate increases the chance of error, keeping the number of layers (Reps) as low as possible mitigates decoherence and gate infidelities. This constraint must be balanced against the need for deeper circuits to capture complex patterns.

- **Error Mitigation:** Various techniques exist to combat noise without full quantum error correction, such as zero-noise extrapolation, readout-error calibration, or quasi-probability techniques. Although they cannot completely eliminate errors, they can partially restore the fidelity of measured results.

- **Post-selection and Decoding:** For certain noise models, post-processing filters or decoding algorithms can identify and discard or correct corrupted measurement outcomes. Such approaches can be valuable if one has a reliable noise model for the hardware in question.

Ultimately, the success of quantum kernels on actual NISQ hardware will hinge on striking a careful balance between increasing circuit complexity to capture non-linear relationships and limiting depth to remain within realistic hardware noise thresholds.

2. Entanglement vs. Overfitting

Entanglement is a key quantum resource that allows qubits to correlate in ways unachievable by classical systems. In the context of QSVMs, entangling gates enable feature maps to capture intricate, high-order relationships that might be inaccessible to simpler kernels. However, there are two caveats:

- **Risk of Overfitting:** Excessive entanglement or overly flexible feature maps can fit the training data too precisely, especially if the training set is small. This is somewhat analogous to using a high-degree polynomial in classical SVMs: while it may yield low training error, generalisation can suffer.

- **Increased Depth and Noise:** More entangling gates typically require deeper circuits, amplifying the issues of gate errors and decoherence on real hardware. Thus, one must balance the desire for stronger entanglement against the practical reality of noise and the risk of over-parameterisation.

In practice, moderate-depth circuits (e.g. $\mathrm{Reps} = 3$ to $\mathrm{Reps} = 5$) may offer a "sweet spot" for NISQ devices, capturing enough complexity to outperform classical kernels without succumbing to noise or severe overfitting.

3. Scalability and Approximate Kernel Methods

Beyond hardware-level noise, a central challenge for quantum kernels lies in scaling to large datasets. Computing the fidelity for every pair of data points is inherently $\mathcal{O}(N^2)$, which can become impractical for big data. Potential strategies to alleviate this overhead include:

- **Randomised Feature Expansions:** One may sample from the quantum feature map stochastically to create a lower-dimensional embedding, akin to random Fourier features in classical kernel approximation.
- **Nyström Approximations:** Instead of computing a full $\mathcal{O}(N^2)$ kernel matrix, one constructs an approximate kernel using a smaller subset of "landmark" points, thereby reducing computational cost.
- **Core-set or Active Learning Approaches:** By identifying the most informative samples first, one can limit kernel evaluations to a smaller fraction of the training set, mitigating the scaling bottleneck.

4. Hybrid Quantum-Classical Enhancements

While the QSVM eventually uses a purely classical SVM optimiser, future research could explore fully hybrid quantum-classical training loops in which the parameters of the feature map itself are variationally tuned. Under such a scheme, one might iteratively adjust both the quantum circuit's parameters (entangling angles, rotation angles, etc.) and the classical SVM parameters, potentially achieving:

- **Better Noise Resilience:** Variationally learned circuits can adapt to hardware noise by settling on parameter regimes that are less susceptible to error.
- **Richer Embeddings:** Data re-uploading or partial re-introduction of classical features can further increase expressivity without arbitrarily deep circuits.
- **Adaptive Regularisation:** A hybrid training loop could incorporate regularisation terms (in the classical cost function) that penalise overly entangled states, mitigating overfitting in the quantum feature map.

5. Outlook

From a physics standpoint, the key to realising the potential of QSVMs lies in balancing circuit depth, entanglement, and noise management. While the simulations in this work demonstrate the theoretical benefits of quantum kernels, practical quantum devices currently face significant limitations. Ongoing improvements in hardware, alongside parallel progress in error mitigation, approximate kernel techniques, and hybrid quantum-classical protocols, will be essential to harness the full potential of QSVMs for cryptographic classification and beyond.

Appendix G: Code and Implementation

The entire source code used in this research, along with detailed installation instructions and usage guidelines, is available at the public GitHub repository below:

`https://github.com/jamesh46/Quantum-Support-Vector-Machines-Research`

Please refer to the repository README for complete instructions.